Counterfactual Attribution of Crypto Systemic Stress on a Ledger-Specified Causal Graph: Methods and a Cautionary Benchmark

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Abstract

Crypto systemic-risk measures usually infer contagion networks from prices. We study what can be learned when the entity-resolved Ethereum ledger instead specifies the parent sets of a linear structural causal model of entity stress. On this skeleton we define an interventional tail measure (OC-CoVaR) and a counterfactual loss attribution with full abduction of event shocks. Interventional effects are flow-free network multipliers, and stress deviations under intervention do not depend on the abducted shocks. Linear Shapley attribution has a walk form that gives intermediaries negative credit, and exact Shapley values need $\mathbf{2}^{\boldsymbol{p}}$ rather than $\boldsymbol{p!\,p}$ evaluations. Applied to the October 2025 liquidation cascade with public data and open address labels, the network equations explain little hourly stress: out-of-sample $\boldsymbol{R^2}$ is 0.005 against 0.161 for an own-lag model, and adding the network to the own lag gives an out-of-sample squared-error gain of -0.3% $[-2.0\%, 1.4\%]$, where a negative gain means larger error. The largest single channel accounts for 15.2% (upper 95% bound 19.5%) of an entity's realised distress. On the same data, zero-shock abduction raises the Binance-to-Bybit share from 6.3% to 65.3%. The observed graph does not clearly beat price-estimated graphs, edge direction is not distinguishable from reversal, and placebo evidence is mixed. We release the pipeline as a reproducible benchmark.

Keywords: systemic risk, crypto-assets, structural causal model, counterfactual attribution, Shapley value, on-chain data

JEL Classification: G01 , G23 , G28 , C32 , C15

1 Introduction

On 10–11 October 2025 cryptocurrency markets experienced one of the largest deleveraging episodes on record, with on the order of nineteen billion US dollars of leveraged positions liquidated within about thirty-six hours.[1] Post-mortems (Ali 2025; Lim 2026) describe a largely mechanical cascade: a mispricing in a centralised exchange's collateral valuation triggered liquidations, which forced sales and further liquidations. Over-collateralised decentralised-finance (DeFi) lending protocols such as Aave recorded essentially no bad debt, whereas centralised-exchange (CEX) risk engines amplified the shock. The episode left a high-resolution on-chain footprint of liquidation transfers, collateral movements and exchange flows, timestamped to the block.

The standard systemic-risk tools are associational. ΔCoVaR (Adrian and Brunnermeier 2016), marginal expected shortfall and SRISK (Acharya et al. 2017), and connectedness measures (Diebold and Yılmaz 2014; Billio et al. 2012) quantify co-movement in returns, and a survey of causal inference in finance finds that applications around these measures remain associational or Granger-based (Kumar et al. 2025). Where causality has been added, the contagion graph is still estimated from prices (Rigana et al. 2024, 2023; Etesami et al. 2023; Giudici and Parisi 2018). Shapley-value risk attribution (Tarashev et al. 2016) and causal Shapley methods (Heskes et al. 2020; Singal et al. 2021) come closest in spirit, but they do not take an observed transaction ledger as the causal structure of a systemic-risk measure (Section 2). None of these makes the ledger the causal skeleton, and none asks how much of measured stress such a skeleton can explain.

We call the framework VEIN (ledger-specified on-chain causal risk attribution) and make four contributions.

[1] See CoinDesk Research, https://www.coindesk.com/research/market-spotlight-the-19-billion-liquidation-that-shook-crypto.

(i) *Framework.* A ledger-specified structural causal model (SCM) skeleton for crypto systemic risk under assumptions A1–A3 (Section 3), in which the in-neighbours of each entity specify its parent set. We define the interventional On-Chain Causal CoVaR (OC-CoVaR) and a counterfactual loss attribution computed by full abduction. The event shocks are recovered by inverting the structural equations on the observed path, so the factual world reproduces realised stress.

(ii) *Analytic results.* Flow-free bounds and the network-multiplier form (Proposition 2, Corollaries 1–3); abduction invariance of stress deviations (Proposition 3) and a share bound (Corollary 4, applied in Section 5.2); the walk representation of linear Shapley attribution, which gives intermediaries negative credit (Proposition 4); an exact $O(2^p)$ subset algorithm (Algorithm 2); and a certified ranking radius (Lemma 1).

(iii) *A methodological lesson with evidence.* Zero-shock abduction computes shares against a model-implied loss and inflates them. On the same data the Binance-to-Bybit share moves from 6.3% under full abduction to 65.3% under zero-shock abduction. A share is bounded by how much of the target's stress the network term explains.

(iv) *A reproducible empirical benchmark* on free public data (BigQuery `crypto_ethereum` and MIT-licensed label repositories), with uncertainty quantified throughout, and with an out-of-sample comparison against an own-lag baseline.

Hourly stress is better predicted by its own lag than by the network: in-sample R^2 is at most 0.273, and out of sample the network's squared-error gain over an own-lag model is -0.3% (negative means larger error; interval $[-2.0\%, 1.4\%]$). No channel's share exceeds 15.2% (interval upper bound 19.5%). Retail and MakerSky head the

ranking, but their order is unresolved and both routes run through the unresolved retail aggregate. Ranks three to six are not resolved in pairwise order. The observed graph does not clearly beat price-estimated graphs, so the hypothesis (H1, Section 3) that the observed graph is better is not supported, and edge direction is not distinguishable from reversal. The time-shift placebo passes, while the rolling calm-window placebo fails in the October 2025 and FTX episodes. The transmitter-absorber check passes for Binance and fails for Ethena. The ledger structure is auditable but explains limited stress at hourly scale, so attribution has to use full abduction and report uncertainty. Section 3 sets out the framework, Section 4 the data, Section 5 the results and Section 6 the discussion.

2 Related Work and Positioning

Associational and estimated-graph measures. ΔCoVaR (Adrian and Brunnermeier 2016) is the change in system value-at-risk when an institution moves from its median state to distress. The SRISK and MES family (Acharya et al. 2017) and connectedness measures (Diebold and Yılmaz 2014; Billio et al. 2012) describe co-movement rather than mechanism, which is the gap the causal survey of Kumar et al. (2025) also documents. Causal-NECO VaR (Rigana et al. 2024) learns the graph with the PC-stable algorithm (Colombo and Maathuis 2014) and computes a Gaussian VaR from the resulting structural model (Rigana et al. 2023; Spirtes et al. 2000). It operates at Pearl Levels 1–2 on foreign-exchange data and estimates its graph. The time-varying directed-information graph of Etesami et al. (2023) is the nearest crypto application. Directed information generalises Granger causality (Granger 1969) but is not a Pearl SCM with do-operator semantics, and both approaches are built from returns. CoRisk (Giudici and Parisi 2018) decomposes risk on a correlation network. Crypto composite indices (Farzulla and Maksakov 2026) and TradFi–DeFi mappings (Aufiero et al.

2025) either estimate networks from prices or describe the on-chain graph without a causal model.

Shapley attribution. Tarashev et al. (2016) allocate system-wide risk to institutions with the Shapley value, and Drehmann and Tarashev (2013) measure the systemic importance of interconnected banks. These allocations use exposures or portfolio losses rather than a causal model of propagation. Causal Shapley values (Heskes et al. 2020) respect a known causal graph, and Singal et al. (2021) give a recursive Shapley attribution for flows in graphical models. Both explain model outputs or flows. VEIN attributes realised systemic loss to interventions on a ledger-specified graph.

Contagion and on-chain data. Network models of default propagation (Eisenberg and Noe 2001; Acemoglu et al. 2015; Glasserman and Young 2016) motivate the network as the object of study. VEIN uses reduced-form linear propagation instead of a clearing mechanism. On-chain data have been used to study Tether issuance (Griffin and Shams 2020), arbitrage and capital flows (Makarov and Schoar 2020), the Terra/Luna run (Liu et al. 2023), DeFi liquidations (Qin et al. 2021), and prediction and forensics (Akcora et al. 2018; Meiklejohn et al. 2013; Weber et al. 2019). Recent crypto work is descriptive or price-based: Sankaewtong et al. (2026) trace USDC depeg contagion through transactions without interventional semantics, Zhang et al. (2026) use correlations of total-value-locked series, and counterfactual analysis has so far reached credit-portfolio losses (Xu 2025). None treats the ledger graph as a structural causal model for a systemic-risk measure. To our knowledge, no published work combines an observed, ledger-specified graph as the parent set of a structural causal model with a counterfactual Shapley attribution of systemic loss. The paper also measures, out of sample against an own-lag baseline, how much ledger-specified propagation explains.

3 The VEIN Framework

3.1 Structural causal model and identification

Let $G = (V, E)$ be a directed, timestamped, dollar-weighted graph of resolved entities (exchanges, lending protocols, stablecoin issuers, custodians and an aggregate retail node) whose edges are observed on-chain flows. For entity i and period t we track a stress state $S_{i,t} \in \mathbb{R}$, oriented so that larger means more distressed, the flow $F_{p \to i,t} \geq 0$, and an exogenous shock $U_{i,t}$. The structural equation is

$$S_{i,t} \;=\; f_i\Big(S_{\mathrm{pa}(i),\,t-1},\; F_{\mathrm{pa}(i)\to i,\,t},\; U_{i,t}\Big), \qquad \mathrm{pa}(i) \;=\; \{\, p \in V : (p \to i) \in E \,\}. \tag{1}$$

Under assumptions A1–A3 below, the in-neighbours of i specify its parent set $\mathrm{pa}(i)$ from the ledger, so only the propagation function f_i requires estimation. We use a ridge-regularised linear specification,

$$S_{i,t} \;=\; b_i + \sum_{p \in \mathrm{pa}(i)} a_{ip}\, S_{p,t-1} + \sum_{p \in \mathrm{pa}(i)} c_{ip}\, \log\!\big(1 + F_{p\to i,t}\big) + U_{i,t}, \tag{2}$$

with parameters estimated on a pre-crisis window; the fitted residuals furnish the shock distribution. The equation has no own lag by design: the parent set is the ledger in-neighbourhood. Section 5.1 measures what this omission costs.

Every parent enters at lag $t-1$ in the stress term, so the system is well posed even when G has cycles.

Proposition 1 (Well-posedness of the lagged system) *Given $S_{\cdot,-1}$ and a shock path, the trajectory generated by (2) is uniquely determined by forward iteration, even when G contains directed cycles.*

Proof At step t the right-hand side of (2) references only stress at $t-1$ and exogenous quantities at t. Induction on t gives existence and uniqueness. □

Identification rests on three assumptions. *Temporal precedence* (A1): a flow $F_{i\to j,t}$ precedes any effect on j at $t' > t$. Block timestamps make this observable, but they show only that a transfer could not be an effect of later stress, not that it is the operative cause. *No unobserved common cause for flow edges* (A2): given the graph up to t, an edge $i \to j$ reflects i's direct action on j. This is the strongest assumption, because the ledger does not record off-chain decisions such as a CEX-internal pricing engine, an oracle update or coordinated withdrawals, any of which can drive both a flow and downstream stress. Section 5.7 reports robustness values that bound but do not remove this threat. *Monotone propagation along flow direction* (A3): distress at i affects j only through realised flow paths from i to j. A3 carries the empirical content; edge reversal (Section 3.5) is a diagnostic of it, not an identification test.

3.2 Interventional and counterfactual risk measures

Over an event window $\mathcal{T}$ let $L_j = \sum_{t\in\mathcal{T}} \max(S_{j,t}, 0)$. The interventional measure applies Pearl's do-operator to (1): entity i is forced to a distressed level s^*, its parents are severed, and the system is propagated forward. On-Chain Causal CoVaR is

$$\text{OC-CoVaR}_q\big(j \,\big|\, \text{do}(S_i = s^*)\big) \;=\; \text{VaR}_q\Big(L_j \,\Big|\, \text{do}(S_i = s^*)\Big), \tag{3}$$

estimated by Monte-Carlo propagation. Each simulated shock path is a circular block draw of joint fitted-residual rows with 24-hour blocks, so all entities share the drawn hours (Section 3.4). The marginal contribution contrasts the distressed intervention

with the median state,

$$\Delta\text{OC-CoVaR}_q(j \mid i) \;=\; \text{OC-CoVaR}_q\big(j \mid \text{do}(S_i = s^*)\big) - \text{OC-CoVaR}_q\big(j \mid \text{do}(S_i = 0)\big), \tag{4}$$

with s^* the 0.95 quantile of i's fit-window stress (the event window is excluded). Entities are ranked by exported tail risk $\mathcal{E}(i) = \sum_{j \neq i} \max(\Delta\text{OC-CoVaR}_q(j \mid i), 0)$.

The counterfactual measure sits one level higher. Following the abduction–action–prediction procedure (Pearl 2009, 2019), we replace i's realised path by its pre-crisis level S_i^{pre}, hold the other entities' abduced shocks fixed, and predict the change in j's loss. With L_j^{fact} the loss in the reference (factual) world,

$$\Delta_i^{CF}(j) \;=\; L_j^{\text{fact}} - L_j^{\text{do}(S_i = S_i^{\text{pre}})}, \qquad \text{share}_i(j) \;=\; \frac{\Delta_i^{CF}(j)}{L_j^{\text{fact}}}. \tag{5}$$

The event shocks are obtained by inverting the fitted equations on the observed path. For $t \in \mathcal{T}$,

$$\hat{U}_{j,t} \;=\; S_{j,t} - b_j - \sum_p a_{jp}\, S_{p,t-1} - \sum_p c_{jp} \log\big(1 + F_{p \to j,t}\big), \tag{6}$$

which includes j's own idiosyncratic shocks. With these shocks the factual path reproduces the observed path to machine precision, and L_j^{fact} is j's realised hinge loss. The attribution is model-relative: it holds the other entities' abduced shocks fixed rather than solving the cross-world equilibrium.

Remark 1 (Zero-shock abduction) Setting $\hat{U} = 0$ on the event window gives a model-implied factual path. By Proposition 3 the deviation r is identical under both abductions, so the numerator differs only through the hinge. The denominator L_j^{fact}, however, shrinks to the part of stress the model generates without shocks, which pushes shares towards one. Section 5.2 quantifies this.

Algorithm 1 VEIN end-to-end pipeline

1: $G \leftarrow$ ResolveEntities(transfers, labels) ▷ Tier-0 sweeps and label matching, Section 4
2: Fit $\hat{\theta}_i$ by ridge regression for each entity i on its pre-event window
3: ΔOC-CoVaR$_q(j \mid i)$ by MonteCarloPropagate under do$(S_i = s^*)$
4: Abduct (full: invert the structural equations on the observed event path), then $\Delta_i^{CF}(j)$ by Predict under do$(S_i = S_i^{\text{pre}})$
5: Joint circular-block pairs bootstrap and Monte-Carlo repeats (Section 3.4)
6: Diagnostics: G versus G^{rev}, G^{sym} and price-estimated graphs; placebos
7: **Output:** OC-CoVaR ranking, counterfactual attributions, falsification and validity battery

3.3 Analytic properties and exact attribution

Let $n = |V|$ and let $A \in \mathbb{R}^{n \times n}$ satisfy $A_{jp} = a_{jp}$ for $p \in \text{pa}(j)$ and $A_{jp} = 0$ otherwise. Over the window $t = 0, \ldots, T-1$, with a fixed initial state, (2) is the linear system

$$S_t = A\, S_{t-1} + \kappa_t, \tag{7}$$

where $\kappa_{j,t} = b_j + \sum_p c_{jp} \log(1 + F_{p \to j,t}) + U_{j,t}$ collects the intercepts, flows and shocks. For an entity i define the *impulse* $\tau_0^{(i)} = e_i$ and, for $t \geq 1$, $\tau_t^{(i)} = A\tau_{t-1}^{(i)}$ with its i-th coordinate reset to one. Write $\tau_{ji}(t)$ for the j-th coordinate. The impulse is the per-unit response of j at t to holding i at a unit distressed level from the start of the window. Expanding the recursion shows that $\tau_{ji}(t)$ is a sum over walks from i to j of length at most t, each walk weighted by the product of its coefficients and not revisiting i. In particular $\tau_{ji}(t) \neq 0$ requires a directed path from i to j of length at most t.

Proposition 2 (Flow-free bounds on the intervention) *Fix i, T and $s > 0$. Let $S^{(s)}$ be the window path under* do$(S_{i,t} = s,\ t < T)$ *and $S^{(0)}$ the path under* do$(S_{i,t} = 0,\ t < T)$*, both with the same initial state and the same inputs κ. Then:*

(a) $S_{j,t}^{(s)} - S_{j,t}^{(0)} = s\,\tau_{ji}(t)$ *for all j and $t < T$, for every realisation of κ;*

(b) for every $q \in (0,1)$, every law of κ, and every flow path,

$$\sum_{t<T} \min\big(s\tau_{ji}(t), 0\big) \;\le\; \Delta OC\text{-}CoVaR_q(j \,|\, i) \;\le\; \sum_{t<T} \max\big(s\tau_{ji}(t), 0\big), \tag{8}$$

where $\Delta OC\text{-}CoVaR_q(j \,|\, i)$ is defined as in (4) with s in place of s^*, and the bounds also hold for the Monte-Carlo estimator when both interventions use the same shock draws.

Corollary 1 (Per-period multiplier) *For $t < T$, $\mathrm{VaR}_q(S^{(s)}_{j,t}) - \mathrm{VaR}_q(S^{(0)}_{j,t}) = s\,\tau_{ji}(t)$ for every q and every law of κ (immediate from Proposition 2(a)). The per-period effect is a deterministic network multiplier.*

Corollary 2 (Zero by construction) *If there is no directed path from i to j of length at most $T-1$, then $\tau_{ji}(t) = 0$ for all $t < T$, so $\Delta OC\text{-}CoVaR_q(j \,|\, i) = 0$ for every q and every law of κ. The same holds for the counterfactual attribution, $\Delta_i^{CF}(j) = 0$.*

Corollary 3 (Exported-risk bound and long-run multiplier) *Let $\Lambda_i(T) = \sum_{j\neq i}\sum_{t<T}\max(\tau_{ji}(t), 0)$. Then (i) $\mathcal{E}(i) \le s\,\Lambda_i(T)$, and $\Lambda_i(T) = 0$ implies $\mathcal{E}(i) = 0$; (ii) if $\rho(A_{-i,-i}) < 1$, then $\tau_{\cdot i}(t) \to m^{(i)} := (I - A_{-i,-i})^{-1}A_{-i,i}$ as $t \to \infty$, so $\Lambda_i(T)/T \to \sum_{j\neq i}\max(m^{(i)}_j, 0)$.*

Proposition 2 implies that the interventional effect is a pure network multiplier: it depends on the graph and the coefficients, not on flows, intercepts or the shock law. Proofs are in Appendix C.

Proposition 3 (Abduction invariance of the stress deviation) *Let S^{f} be a reference path that satisfies (7) for entities outside a set K, generated with inputs κ. For $k \in K$ let $p_{k,t}$ be prescribed levels, and define $g_{k,t} = S^{\mathrm{f}}_{k,t} - p_{k,t}$. Let S^K be the path under $\mathrm{do}(S_{k,t} = p_{k,t},\, k \in K)$ with the same initial state and inputs. Then $S^K = S^{\mathrm{f}} - r(K)$, where r is the linear propagation*

$$r_{k,t} = g_{k,t}\ (k \in K), \qquad r_{j,t} = \sum_p a_{jp}\, r_{p,t-1}\ (j \notin K), \qquad r_{\cdot,-1} = 0. \tag{9}$$

Consequently, for the counterfactual $p_{i,t} \equiv S_i^{\mathrm{pre}}$,

$$\Delta_i^{CF}(j) = \sum_{t<T} \Big[\max\left(S_{j,t}^{\mathrm{f}}, 0\right) - \max\left(S_{j,t}^{\mathrm{f}} - r_{j,t}(\{i\}), 0\right) \Big]. \tag{10}$$

The stress deviation r *does not depend on the flows, intercepts or shock values except through the reference levels at the intervened entities. The only nonlinearity in the attribution is the hinge* $\max(\cdot, 0)$.

Corollary 4 (Share bound) *Under the conditions of Proposition 3,* $|\Delta_i^{CF}(j)| \leq \sum_{t<T} |r_{j,t}(\{i\})|$, *hence* $|\mathrm{share}_i(j)| \leq \sum_{t<T} |r_{j,t}(\{i\})| / L_j^{\mathrm{fact}}$.

Proof Each summand of (10) is $h(S_{j,t}^{\mathrm{f}}) - h(S_{j,t}^{\mathrm{f}} - r_{j,t}(\{i\}))$ with $h(y) = \max(y, 0)$. Since h is 1-Lipschitz, each summand has absolute value at most $|r_{j,t}(\{i\})|$. Summing over $t < T$ gives the first bound, and dividing by $L_j^{\mathrm{fact}} > 0$ gives the second. □ □

Corollary 4 is a bound, not an identity. Informally, r_j is generated only by j's network term. When that term explains a small fraction of j's stress (small R_j^2), shares are small relative to the full-abduction L_j^{fact}, whereas under zero-shock abduction the denominator is not comparable.

Proposition 4 (Walk representation and exact linear Shapley attribution) *Let* $P \subseteq V \setminus \{j\}$ *be the set of candidate sources and* $g_{k,t}$ *as in Proposition 3, with* K *replaced by subsets of* P. *Consider the two games on* P:

$$v(K) = \sum_{t<T} r_{j,t}(K), \qquad v_H(K) = \sum_{t<T} \Big[\max(S_{j,t}^{\mathrm{f}}, 0) - \max\left(S_{j,t}^{\mathrm{f}} - r_{j,t}(K), 0\right) \Big],$$

where $r(K)$ *is defined by* (9) *with* K *as the intervened set. Let a* walk *be a sequence* $(x_0, t_0), (x_1, t_0+1), \ldots, (x_\ell, t_0+\ell)$ *with* $x_\ell = j$, $t_0 + \ell < T$, *and each step along an edge of* G; *let* $\omega(w) = g_{x_0,t_0} \prod_s a_{x_{s+1}x_s}$, *let* $I(w) = \{x_1, \ldots, x_{\ell-1}\} \cap P$ *be the distinct candidate sources among the intermediate nodes, and* $m(w) = |I(w)|$. *Let* $\mathcal{W}$ *be the set of walks with* $x_0 \in P$ *that do not revisit* x_0. *Then:*

(a) $v(K) = \sum_{w \in \mathcal{W}} \omega(w)\, \mathbf{1}[x_0 \in K,\ I(w) \cap K = \emptyset]$;

Algorithm 2 Exact subset Shapley attribution

Input: candidate sources P with $p = |P|$; value oracle $v : 2^P \to \mathbb{R}$ (a linear or hinge window propagation)
for each $K \subseteq P$ **do**
    $v(K) \leftarrow$ oracle evaluation (one forward propagation of the window)
end for
for each $k \in P$ **do**
    $\phi_k \leftarrow \sum_{K \subseteq P \setminus \{k\}} \frac{|K|!\,(p-|K|-1)!}{p!} \big(v(K \cup \{k\}) - v(K)\big)$
end for
Output: $\{\phi_k\}_{k \in P}$ with $\sum_k \phi_k = v(P) - v(\emptyset)$

(b) the Shapley value of v is, for $k \in P$,

$$\phi_k(v) = \sum_{w \in \mathcal{W}:\, x_0 = k} \frac{\omega(w)}{1 + m(w)} \;-\; \sum_{w \in \mathcal{W}:\, k \in I(w)} \frac{\omega(w)}{m(w)\,(m(w)+1)}; \tag{11}$$

(c) $\sum_{k \in P} \phi_k(v) = v(P) = \sum_{w \in \mathcal{W}:\, I(w) = \emptyset} \omega(w)$*;*

(d) if $S^{\mathrm{f}}_{j,t} \geq 0$ *and* $S^{\mathrm{f}}_{j,t} - r_{j,t}(K) \geq 0$ *for all* $K \subseteq P$ *and* $t < T$*, then* $v_H = v$*, so (11) gives the exact Shapley values of the hinge game.*

The games are defined through $r(K)$. By Proposition 3, $v(K)$ is the change in the cumulative stress of j under do on K when the reference path satisfies the structural equations at every entity outside K. The second term of (11) is the negative credit that an intermediate candidate source receives on walks that pass through it. A walk that reaches j through a second candidate source contributes positively to the originating source and negatively to the intermediary. The games here let unselected entities propagate, as in (9). The Bybit Shapley values in Section 5.2 use a different game, in which every non-candidate entity is held at its pre-crisis level, so Proposition 4 does not describe them. The walk representation is verified numerically on synthetic paths only (Appendix D). Proposition 4(d) gives the closed form of the hinge game whenever the hinge is inactive.

Shapley's definition averages marginal contributions over the $p!$ orderings of the players (Shapley 1953). Algorithm 2 evaluates the game once on each of the 2^p subsets and weights the marginals by the number of orderings that place each coalition first. It

is exact and uses no sampling. The number of oracle calls falls from $p!\,p$ for permutation enumeration to 2^p. Each call costs $O(T(|V|+|E|))$ by forward propagation. For $p = 13$ the saving is 8,192 calls against 80,951,270,400 permutation evaluations, which is the comparison used in Appendix D.

3.4 Uncertainty quantification

(a) *Monte Carlo error.* 20 independent repeats of 500 simulations are used. The standard error at the point setting (2,000 paths) is the repeat standard deviation scaled by $\sqrt{500/2{,}000}$. Monte-Carlo error is negligible against the bootstrap widths reported below.

(b) *Coefficient uncertainty.* We use a joint circular-block pairs bootstrap of fit-window rows. All entities share the drawn hours, which preserves cross-sectional dependence. The block length is 24 hours and $B = 300$. Each replicate refits the ridge equations with the same penalty and draws its shock pool from the residuals of its own drawn rows, with 500 simulations per replicate. The sample quantile s^* is held fixed, so its sampling error is not included; this is a limitation. The outputs are percentile intervals for exported risk, percentile rank intervals, P(top-1) and P(top-3), pairwise probabilities that i ranks above j, Spearman correlations between replicate and point rankings, share intervals, and P(attribution > 0). Channel intervals are not selection-adjusted: the top ten channels are chosen on the point estimate.

(c) *Paired prediction comparisons.* We use a circular block bootstrap over event hours, with all entities of an hour resampled together, with $B = 2{,}000$ and 24-hour blocks; 8-hour blocks are a sensitivity check. The circular scheme is used because the non-circular moving-block bootstrap (Künsch 1989) gives the end steps less weight, and with only two blocks this can move the interval away from the sample mean. With two 24-hour blocks the intervals are indicative only.

(d) *Explained variance.* We report in-sample R^2 together with an optimism correction, the mean over replicates of (R^2 on the own resample minus R^2 on the original rows), subtracted from the point estimate (Efron 1983). The interval for the corrected value is the percentile interval of the point estimate minus the replicate optimism. Out-of-sample one-step R^2 uses an expanding-window ridge refit before each event hour, $R^2_{OS}(m) = 1 - \mathrm{SSE}(m)/\mathrm{SSE}(\text{expanding mean})$, pooled over entity-hours of z-scored stress, following the out-of-sample convention of Campbell and Thompson (2008). The network gain over the own lag is $1 - \mathrm{SSE}(m)/\mathrm{SSE}(\mathrm{AR})$. The models are an AR(1) own lag, the network equation of (2) without an own lag, and AR plus network.

(e) *Placebo designs.* The structural zero (Corollary 2) is a property, not evidence. The rolling calm-window statistic is the mean |attribution| over connected ordered pairs. Calm windows have the event length, start every 24 hours after the first 72 fit hours, and end at least 24 hours before the event. With n calm windows, $p = (1 + \#\{\text{calm} \geq \text{event}\})/(n+1)$, and the placebo passes if $p \leq 0.10$. The time-shift placebo circularly shifts each entity's fit-window stress by an independent offset in $[72, n_{\text{fit}} - 72]$ hours, refits with the real flows, and uses $B = 100$ replicates. Its p-value has floor $1/(B+1)$ and it passes if $p \leq 0.05$. The constants (24-hour step, 24-hour gap, 72-hour minimum shift, and the thresholds 0.10 and 0.05) are fixed in the validation code rather than tuned. No multiplicity correction is applied.

3.5 Diagnostics: graph source, direction and confounding

To test A3 we construct three variants of G: the true-direction graph, a reversed graph G^{rev}, and a symmetric graph G^{sym}. Each is fitted on the estimation window and compared on one-step-ahead prediction of event-window stress by the stress-weighted root-mean-square error $\mathrm{RMSE} = \sqrt{\sum_{i,t} w_{i,t}(\hat{S}_{i,t} - S_{i,t})^2 / \sum_{i,t} w_{i,t}}$, with

$w_{i,t} = |S_{i,t}| + 0.1$. Because A2 is not directly testable, we report the robustness value of Cinelli and Hazlett (2020). For a coefficient with t-statistic t on d degrees of freedom and $f = |t|/\sqrt{d}$,

$$RV \;=\; \tfrac{1}{2}\left(\sqrt{f^4 + 4f^2} - f^2\right) \in [0, 1] \tag{12}$$

is the share of residual variance a confounder must explain in *both* the parent and the child to drive the coefficient to zero. Baselines are ΔCoVaR (Adrian and Brunnermeier 2016), MES (Acharya et al. 2017), and, for H1, Granger and partial-correlation graphs (Granger 1969; Giudici and Parisi 2018) fed through the same VEIN machinery. H1 (observed versus estimated graph source) and H2 (true, reversed and symmetric direction) are reported as diagnostics with null-hypothesis wording: the null is that the alternative graph is no better, and a result is called inconclusive when its interval includes zero.

4 Data and Entity Resolution

4.1 Sources, samples and entity set

The analysis uses public data only. Flows come from the BigQuery public dataset `bigquery-public-data.crypto_ethereum.token_transfers` (the `crypto_ethereum` token-transfers view), queried with partition pruning, which was verified by dry run. Prices are from CoinGecko and protocol total-value-locked (TVL) series from DefiLlama. The main analysis is the hourly October 2025 run, with FTX and SVB as extended hourly windows. The FTX window has 912 fit rows, 96 event rows, 14 entities and 81 edges; the SVB window has 888, 96, 14 and 85. The named-to-named share of entity-touching flow is 3.0%, 2.1% and 1.6% in the October, FTX and SVB extracts. A daily core run is used only for the analytic checks of Appendix D.

The hourly entity set is Aave, Binance, Bitfinex, Bybit, Ceffu, Coinbase, Copper, Crypto.com, Ethena, HTX, Kraken, Lido, MakerSky, OKX and retail, with the retail

node defined as all unresolved counterparties. The top-k cut is $k = 12$. The full build billed 327 GiB of BigQuery processing, within the free monthly query allowance at the time of writing.

4.2 Labels and Tier-0 deposit sweeps

Address labels come from three sources. The eth-labels repository (commit `d9b21aefeded56f7fce7208d6fa173d128526c1b`, MIT licence) and the etherscan-labels repository (commit `923aba72c7e2d0682f7ae6194b6140bd90668dc9`, MIT licence) are open. The Dune Spellbook CEX address list (commit `6030202f2d6ee595dc568100c62723f26e5edda5`) is licensed under the Business Source License 1.1. It is fetched at the pinned commit at run time, used only to build internal label tables, and not redistributed. The released label table `results/labels_open.csv.gz` contains MIT sources only (24,959 addresses). In total 27,514 addresses map to a single exchange root, after one conflicting address was dropped.

Only wallet-kind labels and hand-curated protocol seeds enter the entity table (3,484 rows). Deposit-kind labels are almost all Bitget (19,027) and Binance (5,014). Using them as entity labels would bias centrality, so they are held out as the Tier-0 validation set. The DeFi protocol seeds are Aave (10), MakerSky (6), Ethena (2) and Lido (1) addresses.

Tier-0 is a deposit-sweep rule, with parameters read from the build record. An address is flagged when it forwards at least $\tau = 0.8$ of its outflow to one exchange's hot wallets within 24 hours, has at most 3 outgoing counterparties, and sweeps at least US$1,000. A tightened rule ($\tau = 0.9$, at most 2 counterparties) is used where the default fails acceptance (FTX). Acceptance requires precision of at least 0.9 on at least 20 overlapping holdout detections. Table B1 in Appendix B reports the results per extract. Precision on the October extract is 1.00 on 423 overlaps, on SVB 31 overlaps,

and on the daily extract 64 overlaps. FTX has only 13 overlaps, below the rule, so the FTX evidence is limited. Recall by volume is at most 4.3%. The holdout is Binance and Bitget deposits, so precision is not validated for Coinbase, Bybit or Kraken. The swept volume in the October extract is US$74.5bn.

Tier-3 label matching of both endpoints collapses sub-wallets to their roots, and the mint/burn sentinel is dropped. Tiers 1 and 2 (a classifier and an embedding) are exploratory and are assessed in Appendix B.

4.3 Stress states

Stress is the equal-weight mean of the z-scored components of each entity (48-hour rolling window, minimum 6 periods). The flow component is $0.6\,z(\text{net-flow ratio}) + 0.4\,z(\text{total flow})$. The components actually used are listed in `final_results.json`. CEX entities and retail use flow only. Aave adds an AAVE drawdown, Ethena adds the USDe peg deviation, MakerSky adds the DAI peg deviation, and Lido uses the stETH/ETH discount and withdrawal-queue redemption. Components are kept only if non-degenerate on the fit window, and all 15 entities pass. The four DeFi nodes carry price-based components, so they are not purely on-chain measurements.

Lido has no fit-window gross flow (US$0.00bn). The only edge involving Lido in the hourly graph is the Lido→Aave edge. Ethena's fit-window gross flow is US$0.10bn. In FTX and SVB, Ethena does not exist and Lido uses the discount component only. Figure 1 shows the USDe volume around the event and the hourly stress states, with the event window shaded.

4.4 Coverage and a data cross-check

Named-to-named (distinct entities) flow is 3.0% of entity-touching panel-token flow in October. Across the six coverage extracts (the hourly, extended-window and daily

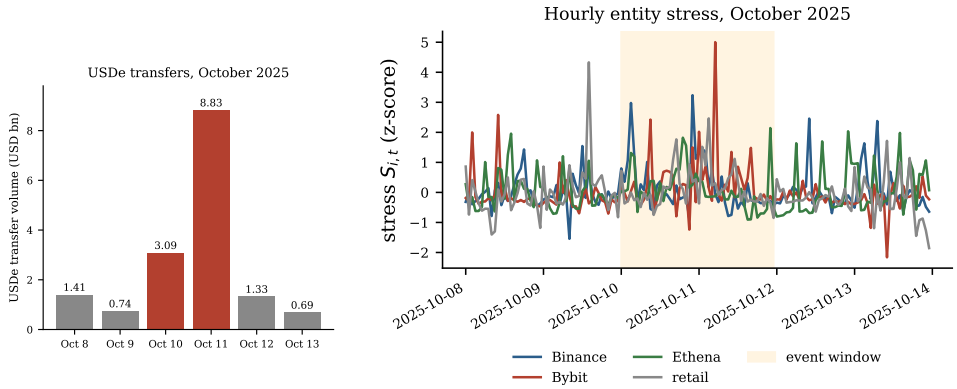


Fig. 1 Left: on-chain USDe transfer volume around the October 2025 cascade (BigQuery `crypto_ethereum` token transfers). Right: hourly stress $S_{i,t}$ of four entities, including the peg component of the Ethena curve (shaded: the 10–11 October event window).

builds) the range is 0.6–3.0%. In the panel that is actually fed to the model, named-to-named flow is 2.8% of US$570bn. The graph is therefore dominated by named-entity–retail edges.

As a cross-check, BigQuery daily USDe volume is compared with an earlier independent extraction. The two match within the 2% tolerance on all six days, and the largest relative difference is 0.06%. The volume peaks at US$3.09bn on 10 October and US$8.83bn on 11 October.

5 Empirical Evaluation

Unless noted, results are for the hourly October 2025 run (600 fitting hours, 48 event hours, 15 entities, 90 directed edges). H1 and H2 are diagnostics, and the numbers come from the `results/*.json` files named in the repository's numbers map. The daily specification is used only for the analytic checks in Appendix D.

5.1 How much stress does the network explain?

Table 1 reports in-sample and out-of-sample explained variance for each entity. In-sample R^2 ranges from 0.000 (Lido) to 0.273 (retail). Optimism-corrected, only retail

Table 1 Explained stress by entity, hourly October 2025. In-sample R^2 and its optimism-corrected value with 95% interval are from the bootstrap; out-of-sample R^2_{OS} is from expanding-window one-step prediction. Rows are sorted by in-sample R^2.

Entity	In-sample R^2	Optimism-corrected [95% CI]	R^2_{OS} AR	R^2_{OS} network	AR+net vs AR (%)
retail	0.273	0.186 [0.077, 0.265]	0.035	0.321	29.4
Kraken	0.140	0.050 [−0.157, 0.158]	0.021	−0.056	−5.8
Coinbase	0.128	0.068 [−0.032, 0.131]	0.022	0.053	11.6
Binance	0.061	−0.008 [−0.089, 0.043]	0.150	0.021	−0.3
Bitfinex	0.052	−0.006 [−0.116, 0.046]	−0.018	−0.203	−16.3
Bybit	0.048	−0.014 [−0.109, 0.037]	−0.040	−0.017	−2.3
OKX	0.042	0.003 [−0.049, 0.038]	−0.004	−0.108	−11.0
Crypto.com	0.023	−0.022 [−0.078, 0.017]	0.007	0.016	1.6
Ceffu	0.021	−0.043 [−0.237, 0.024]	−0.003	−0.015	−1.7
HTX	0.015	−0.031 [−0.098, 0.008]	0.000	−0.120	−12.1
Aave	0.014	−0.041 [−0.150, 0.007]	0.759	0.041	−1.3
Copper	0.011	0.000 [−0.044, 0.016]	0.033	0.023	0.9
MakerSky	0.011	0.001 [−0.034, 0.015]	0.148	−0.027	−0.3
Ethena	0.001	−0.015 [−0.047, −0.001]	0.118	0.020	1.1
Lido	0.000	−0.002 [−0.011, 0.000]	−0.005	0.000	0.0
Pooled (out of sample)	–	–	0.161	0.005	−0.3

stays clearly above zero, at 0.186 [0.077, 0.265]. The Coinbase and Kraken intervals include zero.

Out of sample, the network equation has pooled $R^2_{OS} = 0.005$ [−0.017, 0.019], against 0.161 [0.056, 0.222] for the own lag. The network's squared-error gain over the own lag, $1-\text{SSE}(m)/\text{SSE}(\text{AR})$, is −0.3% [−2.0%, 1.4%] when added to the own lag and −18.6% [−27.1%, −6.2%] when used alone, so the network alone has larger error than the own lag. The large entity-level gains are for retail (29.4%) and Coinbase (11.6%). The other entities lie between −16.3% and 1.6%. Aave has the largest benchmark squared error, about 17.6% of the total, so it carries the most weight in the pooled score. Its own-lag R^2_{OS} is 0.759. Pooled weights follow benchmark squared error, so the equal-weight means are also reported: 0.081 for the own lag and −0.003 for the network. The evidence rests on one window, two 24-hour blocks and one network specification.

5.2 Counterfactual attribution and the abduction lesson

The largest share is 15.2% [13.2%, 19.5%] (MakerSky→retail; the interval is for the maximum over all pairs in each replicate, whereas the channel's own interval is in Table 3). Of the ten largest channels in the point estimate, four end at the retail aggregate, two start there, and four connect two named entities. Table 3 reports the leading channels and Figure 2 draws the top eight.

Among named-to-named channels, Binance→Bybit has a share of 6.3% [2.7%, 8.6%] with $P(\text{attribution} > 0) = 1.00$. Crypto.com→Binance (7.2% [1.3%, 12.5%]) and Binance→OKX (6.7% [0.4%, 11.2%]) have intervals above zero. The Kraken→Bitfinex interval includes zero (6.8% [−9.2%, 12.4%]). The documented Ethena→Aave→Binance mechanism is not detected: Ethena→Aave is −1.1% [−4.1%, 2.3%] and Ethena→Binance is 0.4% [−2.7%, 3.2%].

Table 2 is the abduction lesson. Here v1 and v2 label earlier data builds (v2 is an earlier BigQuery build) and v3 is the build used throughout this paper. On v1, zero-shock abduction gives a Binance→Bybit share of 98.1% ($L_{obs} = 3.49$). On the current data, the same zero-shock abduction gives 65.3% ($L_{obs} = 3.84$), while full abduction gives 6.3% ($L_{obs} = 21.66$). The zero-shock share is inflated because its denominator is a model-implied loss that omits the idiosyncratic shocks: Corollary 4 and the Remark in Section 3.2 explain the mechanism. The zero-shock maximum over pairs is degenerate, with a tiny denominator, and is not quoted. The v1 and v3 rows differ in data as well as in the abduction mode, so only the same-data comparison isolates the abduction effect. Under full abduction the share is similar on the v2 and v3 builds.

For Bybit we also compute a Shapley decomposition of the hinge game over Binance, Coinbase and Bitfinex, by exhaustive enumeration of the orderings. In this game the candidate set is held at its observed stress and every other entity at its pre-crisis level, so only direct channels into Bybit operate. The values are Binance 1.49 [0.53, 2.00], Coinbase 0.12 [−0.09, 0.24] and Bitfinex −0.37 [−0.97, 0.18]. They sum to

Table 2 Abduction lesson for Binance→Bybit. Attribution is the drop in Bybit's loss; the share is attribution divided by L_{obs}, which is the realised hinge loss under full abduction and the model-implied loss under zero-shock abduction.

Data build	Abduction	L_{obs}(Bybit)	Attribution	Share (%)
v1 (earlier extract)	zero-shock	3.49	3.42	98.1
v3 (current build)	zero-shock	3.84	2.51	65.3
v3 (current build)	full	21.66	1.37	6.3
v2 (earlier BigQuery build)	full	21.66	1.45	6.7

Table 3 Leading counterfactual loss-attribution channels $\Delta_i^{CF}(j)$, hourly October 2025. The top ten are chosen on the point estimate, followed by four named-to-named channels. Intervals are not selection-adjusted. The maximum over all pairs is 15.2% [13.2%, 19.5%].

Channel $i \to j$	Type	Attribution	Share (%)	95% CI of share (%)	P(attribution > 0)
MakerSky→retail	→retail	2.19	15.2	[12.2, 18.8]	1.00
Kraken→retail	→retail	1.92	13.3	[6.3, 19.0]	1.00
MakerSky→Coinbase	named→named	1.34	10.9	[7.7, 15.2]	1.00
Coinbase→retail	→retail	1.48	10.3	[5.7, 11.8]	1.00
retail→Coinbase	retail→	1.20	9.8	[5.2, 15.1]	1.00
Bitfinex→retail	→retail	1.31	9.1	[2.7, 15.7]	0.99
retail→OKX	retail→	0.86	8.1	[4.0, 11.2]	1.00
Crypto.com→Binance	named→named	1.36	7.2	[1.3, 12.5]	0.99
Kraken→Bitfinex	named→named	0.61	6.8	[−9.2, 12.4]	0.87
Binance→OKX	named→named	0.72	6.7	[0.4, 11.2]	0.99
Binance→Bybit	named→named	1.37	6.3	[2.7, 8.6]	1.00
Ethena→Aave	named→named	−0.41	−1.1	[−4.1, 2.3]	0.28
Ethena→Binance	named→named	0.08	0.4	[−2.7, 3.2]	0.59
Lido→Aave	named→named	−0.01	0.0	[−0.1, 0.0]	0.12

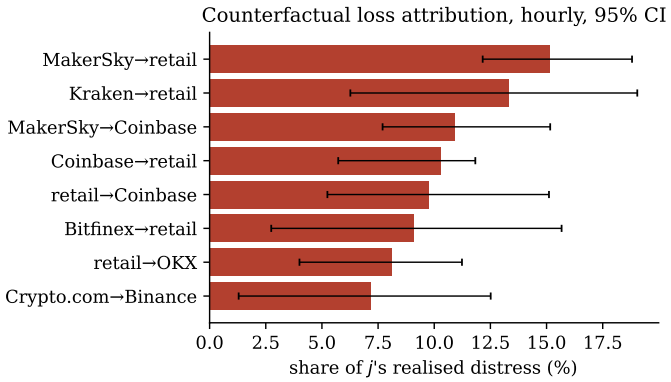


Fig. 2 Top eight counterfactual channels by point share, with 95% bootstrap intervals (300 replicates). The intervals are not selection-adjusted.

the total effect to machine precision (additive residual below 10^{-15}). Only Binance's credit has an interval above zero. This game differs from that of Proposition 4, which lets unselected entities propagate. The negative Bitfinex value is therefore not the intermediary credit of Proposition 4. Its interval includes zero, so we do not trace its source further.

5.3 Systemic ranking with uncertainty

Table 4 reports the ranking and Figure 3 reports the rank intervals and pairwise probabilities. Retail and MakerSky are the top pair: $P(\text{top-3}) = 0.98$ for retail and 0.88 for MakerSky. Their order is unresolved, with $P(\text{retail above MakerSky}) = 0.80$. Ranks three to six are unresolved in pairwise order, with pairwise probabilities between 0.34 and 0.66. Binance is ranked 7th, with rank interval [2, 9]. The five lowest-ranked entities (ranks 11–15) all have $P(\text{top-5}) = 0$. The Spearman correlation between replicate and point rankings is 0.903 [0.793, 0.968].

The retail aggregate matters for the ranking. Without retail, the ranking is led by Binance (17.67), Crypto.com (14.81) and Kraken (14.33), and MakerSky drops out of the top six (point estimates, no interval). Exported-risk levels scale with the number of entities, so only ranks are comparable across runs.

5.4 Graph source and direction diagnostics

Table 5 reports all comparisons. The H1 and H2 labels follow Section 3.5.

H1: observed versus estimated graph.

The weighted RMSE is 1.5545 for the observed graph, 1.5595 for the Granger graph and 1.5599 for the partial-correlation graph ($n = 288$). The primary test uses unweighted squared error with 24-hour blocks. Its paired interval includes zero against both estimated graphs: the mean difference against Granger is -0.0053 [-0.0129, 0.0021], and against partial correlation -0.0031 [-0.0151, 0.0094]. The same holds

Table 4 Exported-risk ranking, hourly October 2025, with bootstrap intervals (300 replicates; the resolution of $P(\cdot)$ is 1/300). Rank intervals are percentile intervals. Spearman correlation between replicate and point rankings: 0.903 [0.793, 0.968].

Rank	Entity	Exported risk	MC s.e.	95% CI of risk	Rank CI	P(top-1)	P(top-3)
1	retail	44.98	0.44	[27.18, 75.19]	[1, 3]	0.77	0.98
2	MakerSky	37.89	0.29	[23.70, 58.08]	[1, 5]	0.17	0.88
3	Bitfinex	26.20	0.32	[8.89, 43.70]	[2, 10]	0.02	0.24
4	Coinbase	26.08	0.28	[10.55, 45.96]	[1, 9]	0.03	0.34
5	Kraken	24.80	0.29	[12.81, 41.64]	[2, 10]	0.00	0.20
6	Crypto.com	23.92	0.31	[8.07, 45.43]	[2, 11]	0.00	0.15
7	Binance	22.35	0.30	[12.06, 47.23]	[2, 9]	0.00	0.17
8	OKX	19.51	0.24	[8.22, 34.16]	[3, 10]	0.01	0.04
9	HTX	13.68	0.25	[3.97, 26.28]	[5, 12]	0.00	0.00
10	Bybit	13.43	0.20	[3.25, 23.69]	[5, 13]	0.00	0.00
11	Copper	4.50	0.13	[1.37, 13.83]	[9, 14]	0.00	0.00
12	Aave	3.38	0.12	[0.08, 13.26]	[9, 15]	0.00	0.00
13	Lido	3.37	0.13	[0.00, 9.26]	[10, 15]	0.00	0.00
14	Ethena	0.70	0.04	[0.04, 7.57]	[11, 15]	0.00	0.00
15	Ceffu	0.55	0.02	[0.00, 1.58]	[13, 15]	0.00	0.00

with 8-hour blocks. In the weighted sensitivity test the Granger comparison is inconclusive, and against partial correlation the interval only just excludes zero. Strictly lagged flows give the same picture (weighted RMSE 1.5496 against 1.5571). The non-flow target and the daily H1 are omitted, since they share the target and have low power. We conclude that H1 is not supported. The observed graph is not clearly better than the estimated graphs.

H2: direction.

On the October run the true, reversed and symmetric graphs have RMSE 1.7841, 1.7871 and 1.7862. The reversed-minus-true interval is [−0.0006, 0.0093], which includes zero, as does the symmetric-minus-true interval (Table 5). In the intraday A3 window, reversal is not distinguishable from the true graph (interval [−0.0035, 0.0166]). The symmetric graph is worse than the true graph, with an interval [0.0003, 0.0250] whose lower bound is barely above zero. Table 5 contains ten uncorrected

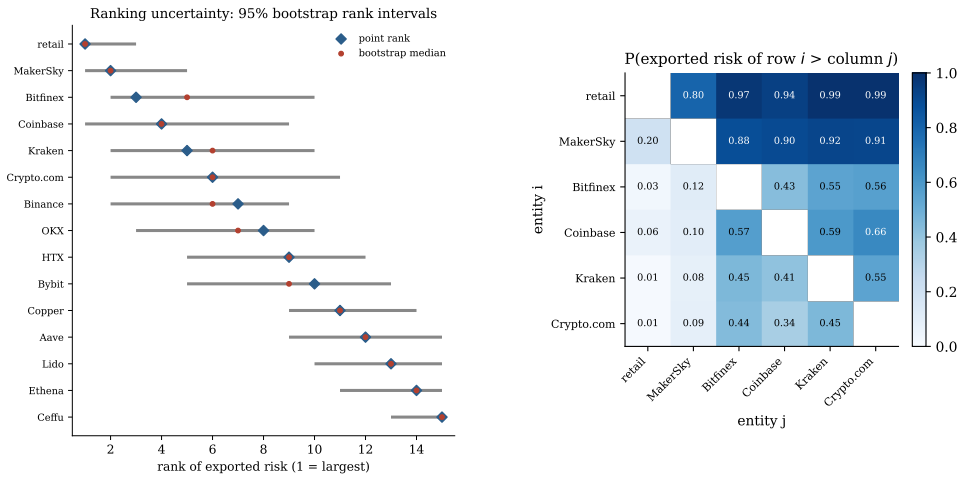


Fig. 3 Left: rank intervals by entity. Right: pairwise order probabilities among the top six entities. Both use 300 refits.

Table 5 Graph-source and direction comparisons. Mean difference is alternative minus base in squared error per entity-hour cell (positive values favour the base). The RMSE columns are stress-weighted, whereas the mean difference and its interval use unweighted squared error, so their signs can differ. Base is the observed graph for H1 and the true-direction graph otherwise. Status is the bootstrap verdict: base better (interval entirely above zero), alternative better (entirely below zero), or inconclusive.

Comparison	Window	n	RMSE base	RMSE alt	Mean diff	95% CI	Status
H1: observed vs Granger	Oct 2025	288	1.5545	1.5595	−0.0053	[−0.0129, 0.0021]	inconclusive
H1: observed vs partial correlation	Oct 2025	288	1.5545	1.5599	−0.0031	[−0.0151, 0.0094]	inconclusive
Reversed vs true	Oct 2025	720	1.7841	1.7871	0.0042	[−0.0006, 0.0093]	inconclusive
Symmetric vs true	Oct 2025	720	1.7841	1.7862	0.0029	[−0.0016, 0.0075]	inconclusive
Reversed vs true	Intraday A3	720	1.8022	1.8036	0.0066	[−0.0035, 0.0166]	inconclusive
Symmetric vs true	Intraday A3	720	1.8022	1.8097	0.0127	[0.0003, 0.0250]	base better
Reversed vs true	FTX, Nov 2022	1,344	1.3019	1.2987	−0.0005	[−0.0025, 0.0014]	inconclusive
Symmetric vs true	FTX, Nov 2022	1,344	1.3019	1.2995	−0.0007	[−0.0037, 0.0021]	inconclusive
Reversed vs true	USDC/SVB, Mar 2023	1,344	1.6514	1.6551	0.0033	[−0.0026, 0.0113]	inconclusive
Symmetric vs true	USDC/SVB, Mar 2023	1,344	1.6514	1.6528	0.0024	[−0.0011, 0.0064]	inconclusive

comparisons, and we do not read this one as evidence for direction. For the two earlier episodes the direction results are inconclusive (Section 5.6). Splitting the October run by edge type, reversing the 86 custodial edges gives an RMSE of 1.7905, and reversing the 4 composability edges gives 1.7806 (true graph 1.7841). No intervals are computed for this split, and the composability result rests on four edges. Direction is not distinguishable from reversal at the entity-aggregate hourly level.

5.5 Placebos and external validity

The structural-zero check holds by construction: there are 196 connected and 14 unconnected pairs, and every unconnected pair has zero attribution. This is an implementation check, not evidence. The rolling calm-window placebo gives $p = 0.286$ over 20 windows, so it fails. The time-shift placebo gives $p = 0.0099$ (floor $1/(B+1) = 0.0099$), so it passes ($B = 100$ shifted refits, seed 11). The fitted network carries timing information, but the event-window attribution is not unusual relative to calm windows. The FTX rolling placebo gives $p = 0.188$ (fails) and the SVB rolling placebo gives $p = 0.032$ (passes). These are uncorrected.

External validity is checked by comparing transmitters (Ethena and Binance) with absorbers (Aave and Lido). Pairwise concordance is 0.50 on the point estimate (0.65 averaged over the bootstrap replicates), and the separation is 8.15. The probability that all transmitters rank above all absorbers is 0.14. Binance passes the check, but Ethena fails at rank 14. All four entities have non-degenerate stress, so the check is not vacuous.

5.6 Multi-event validation

We rerun the pipeline on two earlier episodes: the FTX collapse (event window 8–11 November 2022) and the USDC depeg around the Silicon Valley Bank failure (event window 10–13 March 2023). These use the same CEX label set and the hourly specification (Appendix A). Table 6 reports the top exporters, the rolling placebo and the direction verdict.

FTX is the top exporter, with rank interval [1, 3] and $P(\text{top-3}) = 1.00$. This is consistent with the failing venue exporting stress, but it is descriptive: FTX's stress is flow-based and dominated by its own outflows, and no causal claim is made. Retail is second and Binance third. In the SVB episode retail leads, followed by Binance and Coinbase.

Table 6 Multi-event validation (hourly). Top three exporters with rank interval and P(top-3), the rolling calm-window placebo p-value with the number of windows, and the direction verdict for reversal and symmetry.

Event	Top three exporters (rank CI; P(top-3))	Rolling p (windows)	Reversed	Symmetric
Oct 2025	retail rank [1, 3], $P = 0.98$; MakerSky rank [1, 5], $P = 0.88$; Bitfinex rank [2, 10], $P = 0.24$	0.286 (20)	inconclusive	inconclusive
FTX, Nov 2022	FTX rank [1, 3], $P = 1.00$; retail rank [1, 3], $P = 0.98$; Binance rank [2, 6], $P = 0.64$	0.188 (31)	inconclusive	inconclusive
USDC/SVB, Mar 2023	retail rank [1, 2], $P = 1.00$; Binance rank [1, 5], $P = 0.85$; Coinbase rank [2, 7], $P = 0.53$	0.032 (30)	inconclusive	inconclusive

Direction is inconclusive in both episodes. In FTX the reversed graph has the lowest RMSE, but the interval includes zero. In SVB the true-direction graph has the lowest RMSE, the symmetric graph is close behind, and the reversed-minus-true interval includes zero. These comparisons are low-powered. The earlier episodes have 912 and 888 fit rows, and the daily robustness runs have 38 and 37 fit rows against a required minimum of 125. The daily runs are therefore underpowered and are not reported.

5.7 Robustness

The top-k, edge-dropping, latent-shock and coefficient-noise checks are point comparisons without bootstrap intervals (`topk_sensitivity.json`, `revision*.json`). The top-k check compares the main specification with $k = 10$ and $k = 15$. The Spearman correlation over the common entities is 0.984 ($k = 10$) and 0.986 ($k = 15$), and the top-three overlap with the main run is 3 and 3 of three entities. The top channel, MakerSky→retail, has share 15.6%, 15.2% and 16.4% at $k = 10$, 12 and 15.

For confounding, the robustness values range from 0.089 to 0.231 across the 15 edges with the largest t-statistics of the 90 in the hourly graph, with MakerSky→retail at 0.231. These values are moderate, not large. Dropping the 22 lowest-RV edges leaves the Spearman correlation at 0.989. A latent common shock of strength 0.3 and 0.6 gives Spearman correlations of 0.986 and 0.904, with a mean top-three overlap fraction of

0.67. Coefficient noise (15% relative standard deviation) changes the Binance→Bybit share by a standard deviation of 0.65 percentage points, and the share stays positive in every draw. The bootstrap width dominates.

Under the alternative resolution rule, the Spearman correlation with the main ranking is 0.714 over 6 common entities (p=0.111), which is not established. For the price-based baseline, the Spearman correlation between the OC-CoVaR ranking and ΔCoVaR over six mapped entities is 0.371 (n=6). This is not distinguishable from zero, and is an illustration only.

6 Discussion

What the ledger buys.

The observed transaction graph is auditable. Every edge maps to inspectable transfers, structural zeros are exact, and attributions are reproducible from public data. It is not sufficient. At hourly scale the own lag and idiosyncratic shocks account for more of stress than the network does, and the ledger-specified network adds nothing detectable out of sample.

Implications for supervisors and risk teams.

Ledger graphs are useful as an audit layer and a hypothesis generator, not as a standalone contagion estimate. Any "share of distress" statement should use full abduction, because zero-shock abduction inflates the Binance→Bybit share from 6.3% to 65.3% on the same data. Explained-variance diagnostics should be reported next to attributions, and rank uncertainty should be reported rather than point ranks.

Limits.

CEX opacity is the main threat to A2: the proximate cause, an exchange's pricing engine, is off-chain, and the robustness values bound but cannot remove this threat. Label coverage is limited: named-to-named flow is about 3.0% of entity-touching flow,

volume recall of the Tier-0 rule is at most 4.3%, and validation covers only Binance and Bitget deposits. The retail aggregate dominates both the ranking and the channels. The model is linear and has no own lag, and s^* is held fixed in the bootstrap. There are three events, and the tests are uncorrected for multiplicity. The DeFi nodes use price-based components.

Extensions.

Own-lag or regime-dependent structural equations, nonlinear clearing, intraday horizons, and better CEX resolution are natural next steps.

7 Conclusion

We treat the entity-resolved on-chain ledger as a candidate structural skeleton. We give an interventional measure, OC-CoVaR, and a counterfactual loss attribution with full abduction. We prove flow-free bounds, the network-multiplier form, abduction invariance of the stress deviation, a share bound, the walk representation of the linear Shapley attribution and an exact subset algorithm. On the October 2025 event the framework is auditable but explains little hourly stress out of sample, does not clearly beat price-estimated graphs, and cannot identify edge direction. Its share-based statements are reliable only under full abduction and with explained-variance diagnostics. The benchmark is released so that these conclusions can be checked on free public data.

Statements and Declarations. Data availability. All code, query definitions and derived result files (`results/*.json`) underlying every number, table and figure are available at https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk. Raw transfers are not redistributed. They are regenerated from the public BigQuery dataset `bigquery-public-data.crypto_ethereum` (token

transfers), which can be queried in the free BigQuery sandbox; the full build processed 327 GiB. Address labels come from the eth-labels (commit `d9b21aefeded56f7fce7208d6fa173d128526c1b`) and etherscan-labels (commit `923aba72c7e2d0682f7ae6194b6140bd90668dc9`) repositories, both MIT-licensed, and a label table built from these two sources alone is included. The Dune Spellbook CEX address list (commit `6030202f2d6ee595dc568100c62723f26e5edda5`) is licensed under the Business Source License 1.1; it is fetched at the pinned commit at run time, used only to build internal label tables, and is not redistributed. Prices are from CoinGecko and protocol TVL from DefiLlama. No restricted or proprietary data were used.

Code availability. Code is in the repository named in the data availability statement.

Ethics approval and consent to participate. Not applicable: this study uses only public, aggregated on-chain and market data and involves no human subjects.

Consent for publication. Not applicable.

Competing interests. The author declares no competing interests.

Funding. No external funding supported this work.

Authors' contributions. Q.-V.D. conceived the study, developed the VEIN framework and its implementation, performed the data collection and empirical analysis, and wrote the manuscript.

Appendix A Implementation details and backtests

The pipeline is implemented in Python, with on-chain data queried from BigQuery and cached by query hash. Structural equations (2) are fitted by ridge regression with penalty $\lambda = 1$. The deposit-sweep threshold is $\tau = 0.8$, the VaR level is $q = 0.95$, and the seeds are 7 for the point estimate, 20,251,010 for the bootstrap, 20,251,011 for the attribution, and 11 for the time-shift placebo. The Monte-Carlo point estimate uses 2,000 simulations, full abduction of the event-window shocks, joint residual rows and

Table B1 Tier-0 validation against the holdout deposit labels. Recall by count and by volume and the precision are for the chosen rule.

Extract	Rule	τ	Holdout	Overlap	Wrong root	Precision	Recall (count)	Recall (volume)
Oct 2025 (hourly)	default	0.8	25,080	423	0	1.00	68.3	0.4
FTX (hourly)	tightened	0.9	25,080	13	0	1.00	17.1	4.3
USDC/SVB (hourly)	default	0.8	25,080	31	0	1.00	34.1	3.2
Daily core	default	0.8	25,080	64	0	1.00	55.2	0.1

the sample quantile s^* = the 0.95 quantile of the fit window only (600 hours). Each bootstrap replicate uses 500 simulations. The coefficient-perturbation checks multiply every stress and flow coefficient by $1+\eta$ with $\eta \sim N(0, 0.15^2)$. The H1 bootstrap uses $B = 2{,}000$ circular blocks.

Appendix B Entity-resolution validation

Table B1 reports the Tier-0 validation per extract. The Tier-0 rule is the deposit-sweep rule of Section 4.2, with the tightened parameters used only for FTX.

The Tier-1 classifier and the embedding are validated against CEX-labelled addresses in the USDe window (8–14 October 2025). After the mint/burn sentinel is excluded, the window has 7,019 addresses, of which 41 are CEX-labelled (base rate 0.6%). The held-out ROC-AUC is 0.89, and the average precision is 0.17. At the best-F_1 threshold the classifier's precision and recall by address count are 0.13 and 0.58. By USD volume, precision is 0.92 and recall is only 0.15. The Tier-2 cluster homogeneity is 0.10. The ground truth is small and deposit-dominated, so these numbers are a weak check. Tier-3 label matching therefore remains the production resolver.

Appendix C Proofs

Let $h(y) = \max(y, 0)$, which is nondecreasing and 1-Lipschitz, so $\min(\delta, 0) \le h(x+\delta) - h(x) \le \max(\delta, 0)$.

Proof of Proposition 2. (a) Let $D_t = S_t^{(s)} - S_t^{(0)}$. Both paths share the initial state and the inputs κ, so $D_{-1} = 0$. For $j \neq i$, $D_{j,t} = \sum_p a_{jp} D_{p,t-1}$. For i, $D_{i,t} = s$ for $t < T$. We show $D_t = s\tau_t$ by induction. At $t = 0$, $D_0 = se_i = s\tau_0$. For $t \geq 1$, the $j \neq i$ coordinates satisfy $D_{j,t} = s\sum_p a_{jp}\tau_p(t-1) = s\,(A\tau_{t-1})_j = s\,\tau_j(t)$, since the reset affects only the i-th coordinate, and $D_{i,t} = s = s\,\tau_i(t)$.

(b) For a fixed realisation let $x_t = S_{j,t}^{(0)}$. Then $L_j^{(s)} - L_j^{(0)} = \sum_{t<T}[h(x_t + s\tau_{ji}(t)) - h(x_t)]$, and each summand lies in $[\min(s\tau_{ji}(t), 0), \max(s\tau_{ji}(t), 0)]$. The sums ℓ and u of these endpoints are deterministic, so $L_j^{(0)} + \ell \leq L_j^{(s)} \leq L_j^{(0)} + u$ pointwise. Quantiles are monotone and translation-equivariant, which gives $\mathrm{VaR}_q(L_j^{(0)}) + \ell \leq \mathrm{VaR}_q(L_j^{(s)}) \leq \mathrm{VaR}_q(L_j^{(0)}) + u$, and the same holds for the Monte-Carlo estimator with shared draws. □

Proof of Corollary 2. The induction in the text shows that $\tau_{ji}(t) \neq 0$ requires a directed path from i to j of length at most t. Without such a path, $\tau_{ji}(t) = 0$ on the window and Corollary 1 gives $\Delta\text{OC-CoVaR}_q(j \mid i) = 0$. For the attribution, $r_j \equiv 0$ in Proposition 3 with $K = \{i\}$, so (10) gives $\Delta_i^{CF}(j) = 0$. □

Proof of Corollary 3. (i) The upper bound in Proposition 2(b) is non-negative for $s > 0$, so $\max(\Delta\text{OC-CoVaR}_q(j|i), 0) \leq s\sum_t \max(\tau_{ji}(t), 0)$. Summing over $j \neq i$ gives $\mathcal{E}(i) \leq s\Lambda_i(T)$. So $\Lambda_i(T) = 0$ forces $\mathcal{E}(i) = 0$. (ii) For $t \geq 1$, $x_t = \tau_{\cdot i}(t)$ restricted to $V \setminus \{i\}$ satisfies $x_t = A'x_{t-1} + b$, with $A' = A_{-i,-i}$, $b = A_{-i,i}$, $x_0 = 0$. With $m = (I - A')^{-1}b$ and $\rho(A') < 1$, we have $x_t - m = A'^t(x_0 - m) \to 0$. The Cesàro mean of the convergent sequence $\max(x_t, 0)$ converges to $\max(m, 0)$, which gives the limit of $\Lambda_i(T)/T$. □

Proof of Proposition 3. Let $D = S^K - S^{\mathrm{f}}$. Both paths share the inputs κ and the initial state, so $D_{-1} = 0$. For $k \in K$, $D_{k,t} = p_{k,t} - S_{k,t}^{\mathrm{f}} = -g_{k,t}$. For $j \notin K$, both paths satisfy (7), so $D_{j,t} = \sum_p a_{jp} D_{p,t-1}$. Induction on t shows $D = -r$, since r satisfies the same recursion with the same forcing. Hence $S^K = S^{\mathrm{f}} - r(K)$. Taking

$K = \{i\}$ and $p_{i,t} \equiv S_i^{\text{pre}}$ gives $L_j^{\text{do}} = \sum_t \max(S_{j,t}^{\text{f}} - r_{j,t}(\{i\}), 0)$, and (10) follows from $\Delta_i^{CF}(j) = L_j^{\text{fact}} - L_j^{\text{do}}$. □

Proof of Proposition 4. (a) Unrolling the recursion for r backwards from (j,t), a node in K stops the recursion as a forcing term and a node outside K passes it to its parents. Hence $r_{j,t}(K)$ is the sum of $\omega(w)$ over walks from a forced node $x_0 \in K$ whose intermediate nodes avoid K, which is the indicator in (a). Walks that revisit x_0 contain $x_0 \in K$ among their intermediates and are excluded by the same condition, so restricting to $\mathcal{W}$ changes nothing.

(b) By linearity it suffices to treat $u_w(K) = \mathbf{1}[x_0 \in K,\ I(w) \cap K = \emptyset]$. Let $k = x_0$, $I = I(w)$, $m = |I|$, and let Pre_y be the players preceding y in a uniformly random ordering; the Shapley value of y is its expected marginal contribution. For $y = k$ the marginal is $\mathbf{1}[\text{Pre}_k \cap I = \emptyset]$, and k precedes all of I with probability $1/(m+1)$. For $y \in I$ the marginal is $-\mathbf{1}[k \in \text{Pre}_y,\ \text{Pre}_y \cap I = \emptyset]$. This requires k to be first and y second among $\{k\} \cup I$, which has probability $1/((m+1)m)$. For any other y the game does not depend on y. This gives $\phi_k(u_w) = 1/(1+m)$, $\phi_y(u_w) = -1/(m(m+1))$ for $y \in I$, and zero otherwise. Summing over walks with the weights $\omega(w)$ gives (11).

(c) Summing (11) over $k \in P$ gives $\omega(w)[\frac{1}{1+m} - \frac{1}{m+1}] = 0$ for $m \geq 1$ and $\omega(w)$ for $m = 0$, which matches (a) at $K = P$.

(d) If $S_{j,t}^{\text{f}} \geq 0$ and $S_{j,t}^{\text{f}} - r_{j,t}(K) \geq 0$, then both hinge terms are linear, so $v_H(K) = v(K)$. □

Lemma 1 (Certified ranking radius) *Let $\tilde{A}$ satisfy $|\tilde{a}_{jp} - a_{jp}| \leq \varepsilon |a_{jp}|$ with the same sparsity pattern, and let $\tilde{\Lambda}_i$ be $\Lambda_i(T)$ computed from $\tilde{A}$. Then $|\tilde{\Lambda}_i - \Lambda_i(T)| \leq R_i(\varepsilon)$, where $R_i(\varepsilon) = \sum_{j \neq i} \sum_{t<T} \left[\tau_{ji}^{(1+\varepsilon)|A|}(t) - \tau_{ji}^{|A|}(t)\right]$. If $\Lambda_r - \Lambda_{r+1} > R_r(\varepsilon) + R_{r+1}(\varepsilon)$, the order of the pair is preserved under every such perturbation.*

Proof $\tau_{ji}(t)$ is a sum over walks from i to j of length at most t avoiding i in between. The telescoping bound $|\prod \tilde{a}_e - \prod a_e| \le \prod(1+\varepsilon)|a_e| - \prod |a_e|$ gives the entrywise bound, and h is 1-Lipschitz. □

Appendix D Daily analytic checks and bounds

The numbers come from `results/analytics.json`, which uses the daily SCM with $T = 7$ and 15 entities; the spectral radius is $\rho(A) = 0.569$. B^+ is $\Lambda_i(T)$ per unit s^* (Corollary 3), so the stored exported risk must satisfy $\mathcal{E}(i) \le s^* B^+$, and the implied minimum s^* is $\mathcal{E}/B^+$. The infinite-horizon multiplier is a reference quantity, not a bound. These checks were not run on the hourly coefficients, which are archived in `hourly_scm_export.json`.

There are 0 violations of the bound. 5 entities have $B^+ = 0$ and stored $\mathcal{E} = 0$; the bound is vacuous for them because their coefficients are all zero. This is a one-directional sanity check, not out-of-sample validation. Table D1 lists the bounded entities; the other 5 have $B^+ = 0$. Coefficients are archived to four decimals, so the implied minimum s^* values are approximate.

The certified ranking radius (Lemma 1) applies to the ranking by $\Lambda_i(T)$ under relative coefficient perturbations of size ε. For the top five daily entities (retail, Aave, Lido, MakerSky and Copper), the certified radii for consecutive pairs are 0.047, 0.064, 0.134 and 0.305, and no pair is certified at $\varepsilon = 1$. The certificate does not cover the exported-risk ranking $\mathcal{E}$, flows, shocks, s^* or estimation error.

The closed-form walk decomposition agrees with subset enumeration to within 10^{-15} on a synthetic deviation path (seed 7), and the subset and permutation computations agree to the same precision. Subset enumeration needs 8,192 oracle calls against 80,951,270,400 for permutations at $p = 13$, the retail parent set. The exact subset computation for retail took 0.22 s on one machine.

Table D1 Flow-free bounds on exported risk, daily specification ($T = 7$), bounded entities only.

Entity	Stored $\mathcal{E}$	B^+ per unit s^*	Implied min. s^*	∞-horizon multiplier
retail	8.284	4.414	1.88	0.737
Lido	6.678	3.151	2.12	0.675
Aave	6.457	3.906	1.65	0.682
MakerSky	1.161	1.780	0.65	0.373
Ethena	0.041	0.243	0.17	0.041
Bybit	0.024	0.034	0.70	0.007
Ceffu	0.019	0.037	0.52	0.009
Copper	0.002	0.365	0.00	0.061
Binance	0.001	0.005	0.28	0.001
FalconX	0.000	0.058	0.00	0.014

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